

Group 1 – Cyclist Data Analysis

Group 1 Objective

The Cycling Group will examine changes in cycling activity at treatment streets and provide supporting evidence for the wider parking utilisation analysis, particularly for streets where parking utilisation has decreased.

Group 1 Members

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Sprint 1 Aim

Our aim is to identify a suitable cycling site and provide one simple cycling result to support the wider parking utilisation analysis.

Useful Data Sources

- **Bicycle Volume and Speed** – cycling count data for the before-and-after analysis.
- **Bike Site Number Listing** – used to match counter IDs with street locations and coordinates.
- **streets_spatial.xlsx** – used to confirm treatment roads and street segments.
- **street_segment_qtr_attributes.csv / sites_db.csv** – used to check intervention timing and construction periods.
- **Bicycle Infrastructure Network** – used as supporting information for cycling infrastructure.

1. Site Selection

I reviewed the 96 treatment street segments and compared them with 58 cycling counter locations using GPS coordinates.

Four candidate streets were identified:

Site	Counter	Result
Albert Street	9077	Selected for preliminary analysis
Wellington Street	32493	Insufficient cycling data before the 2015 intervention
Heidelberg Road	40004, 40005	No cycling data for the intended before period
Moorabool Street	34687	No usable cycling records for the required period

Albert Street was selected as the main site because it had the most usable cycling data around the intervention period.

2. Albert Street Data Check

- Counter 9077 is matched to Albert Street segment 9801.
- The segment changed from a painted bike lane to a protected bike lane on 4 June 2020.
- Cycling data is available from 1 January 2019 to 25 December 2022.
- 1,287 of 1,461 expected days contain observations.
- Overall data coverage is approximately 88.1%.
- A major 168-day collection gap occurs from 5 August 2019 to 19 January 2020 in the before-intervention period.
- A further 6-day gap occurs at the end of December 2022.

The large before-period gap is the main limitation of the Albert Street analysis.

Data Quality Evidence

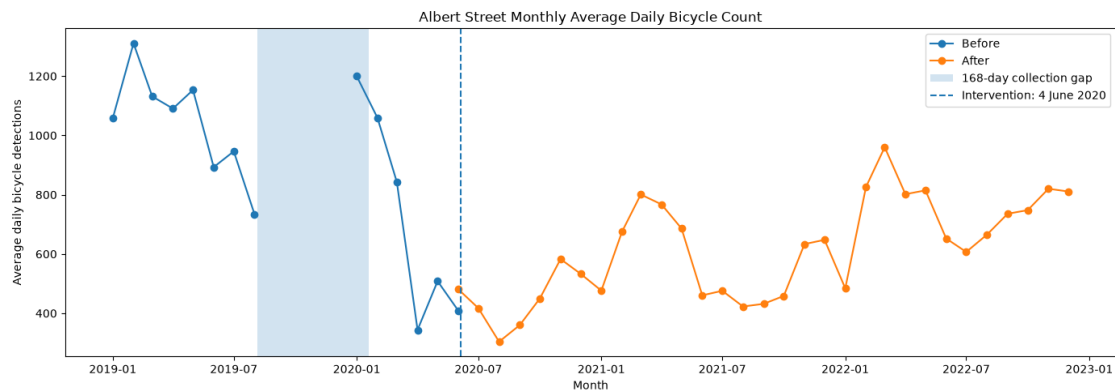


Figure 1. Albert Street counter 9077 data coverage and collection gaps.

3. Duplicate Check

Duplicate-looking records were identified during the initial quality check, so a separate sensitivity test was completed before finalising the analysis.

- 363 exact duplicate rows were removed from the analytical copy.
- Raw source data was kept unchanged.
- Removing exact duplicates changed the estimated before-and-after result only slightly.

Data	Before Average	After Average
Original	939.88	611.14 (-34.98%)
Exact duplicates removed	939.33	610.96 (-34.96%)

The difference was only around 0.02 percentage points, so duplicates did not materially affect the result.

The large collection gap remains the more important data-quality issue.

Duplicate Sensitivity Evidence

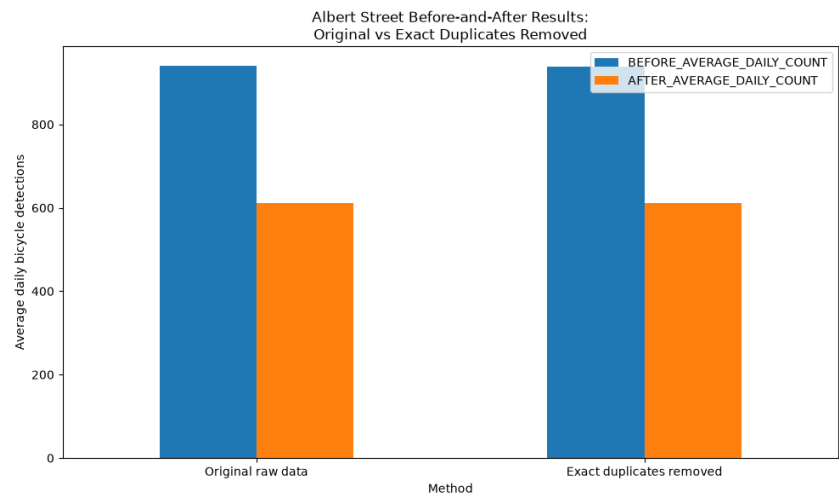


Figure 2. Original versus exact-duplicate-cleaned Albert Street result.

4. Preliminary Before-and-After Analysis

A simple observed-days-only before-and-after comparison was completed for Albert Street.

Method

- Intervention date: 4 June 2020
- Exact duplicates removed from the analytical copy
- Missing days were not treated as zero
- Average daily cycling counts calculated using available observations

Result

Period	Average Daily Cycling Count
Before	939.33
After	610.96

Preliminary percentage change: -34.96%

Average daily cycling counts were approximately 35% lower after the intervention in the available observations.

Result Visual

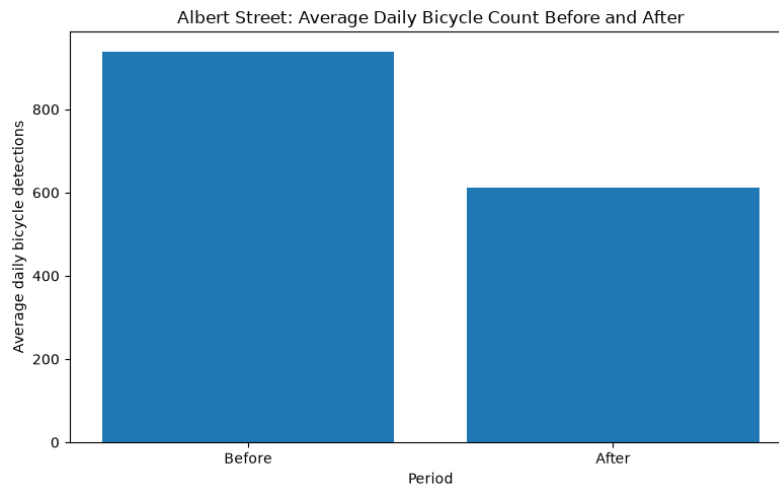


Figure 3. Preliminary Albert Street before-and-after average daily cycling counts.

Interpretation

The result should be treated as preliminary because the before period contains a substantial 168-day collection gap. Seasonal differences, COVID-19 disruptions and other external factors may also have affected cycling activity.

Therefore, the result does not show that the bike-lane intervention caused the observed decrease.

Sprint 1 Finding

Albert Street was the only candidate site with enough cycling data to attempt a preliminary before-and-after comparison.

Average daily cycling counts decreased from approximately 939 before the intervention to 611 after the intervention, a preliminary decrease of approximately 35%.

However, a major 168-day collection gap affects the before period. Duplicate sensitivity testing showed that duplicate removal had almost no effect on the result, so missing data is the more important limitation.

The Albert Street result should therefore be used as exploratory supporting evidence rather than a reliable causal finding.

Next Steps

- Compare the Albert Street cycling result with Team B's parking utilisation findings.
- Finalise the Albert Street visual and short interpretation.
- Clearly document the missing-data limitation.
- Keep Wellington Street, Heidelberg Road and Moorabool Street documented as excluded sites.
- Upload the final notebooks, charts and findings to GitHub after team review.

Documentation

Sprint 1 work includes:

- site and counter matching;
- intervention-date checking;
- cycling data extraction;
- data availability and quality checks;
- duplicate investigation and sensitivity testing;
- preliminary Albert Street before-and-after analysis;
- charts and findings summary.

Main Evidence to Keep

- Albert coverage / collection-gap screenshot
- Duplicate sensitivity screenshot
- Albert before-and-after chart

Backup Evidence

- Candidate site / GPS matching result
- Wellington counter 32493 → segment 5881 verification
- Heidelberg missing historical data output
- Moorabool no-record output
- Detailed duplicate investigation output